

Exp No:12 EMBEDDED C PROGRAM TO CONVERT THE
HEXADECIMAL DATA 0XCFH TO DECIMAL AND DISPLAY THE
DIGITS ON PORTS P0, P1 AND P2

AIM:

To write an Embedded C program to convert Hexadecimal numbers into Decimal and display the digits on ports using Keil software.

PROGRAM:

```
#include<reg51.h>
void main(void)
{
    unsigned char hexa = 0x08;
    unsigned char hundreds,tens,units;

    ACC=hexa;
    B = 10;
    ACC =ACC / B;
    units=B;

    B =10;
    ACC =ACC / B;
    tens = B;
    hundreds=A
    CC;
```

```

P0
=units
;

P1
=tens;

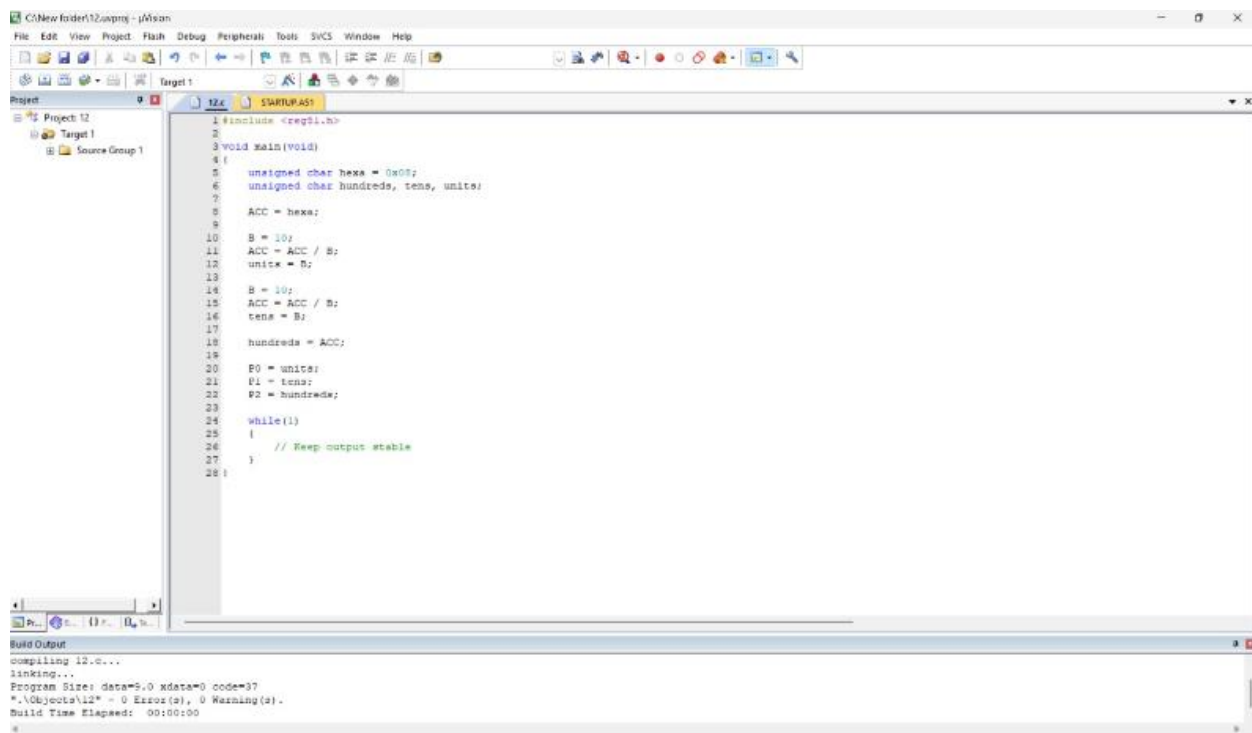
P2 =hundreds;

while(1);

}

```

SIMULATION:



OUTPUT:

The screenshot displays the µVision IDE interface with the following components:

- Registers Window:** Shows the current state of the processor registers. The PC register is at 0x0000, and the ACC register is at 0x0000. The SP register is at 0x0000.
- Code Editor:** Displays the C program for the microcontroller. The program calculates the sum of digits of a hexadecimal number 0x0800.
- Command Window:** Shows the execution progress and an error message: "error 35: undefined line number".
- Call Stack - Locals:** Shows the current function call stack, with the main function at location 0x0000.
- Parallel Port Windows:** Three windows (Parallel Port 0, 1, and 2) are open, showing the current state of the parallel ports. All ports are currently at 0x00.

```
1: void main(void)
2: {
3:     unsigned char hexa = 0x08;
4:     unsigned char hundreds, tens, units;
5:     T:
6:
7:     ACC = hexa;
8:     B = 10;
9:     ACC = ACC / B;
10:    units = B;
11:
12:    B = 10;
13:    ACC = ACC / B;
14:    tens = B;
15:
16:    hundreds = ACC;
17:
18:    P0 = units;
19:    P1 = tens;
20:    P2 = hundreds;
21:
22:    while(1)
23:    {
24:        // Keep output stable
25:    }
26: }
```

Command Window:

```
Running with Code Size Limit: 2K
Load "C:\New Folder\Objects\12"
SS \12\7
*** error 35: undefined line number
```

Call Stack - Locals:

Name	Location/Value	Type
MAIN	C:0x0800	